

A view of 0 or 1 elements: `views::maybe`

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Abstract: This paper proposes `views::maybe` a range adaptor that produces a view with cardinality 0 or 1 which adapts nullable types such as `std::optional` and pointer to object types.

§ 1 Changes

§ 1.1 Changes since R5

- Fix reversed before/after table entry
- Update to match C++20 style [[N4849](#)] and changes in Ranges since [[P0896R3](#)]
- `size` is now `size_t`, like other ranges are also
- add synopsis for adding to `<ranges>` header
- Wording clean up, formatting, typesetting
- Add implementation notes and references

§ 1.2 Changes since R4

- Use `std::unwrap_reference`
- Remove conditional `noexceptness`
- Adopted the great concept renaming

§ 1.3 Changes since R3

- Always Capture
- Support `reference_wrapper`

§ 1.4 Changes since R2

- Reflects current code as reviewed
- Nullable concept specification

Remove Readable as part of the specification, use the useful requirements from Readable

- Wording for views::maybe as proposed
- Appendix A: wording for a view_maybe that always captures

§ 1.5 Changes since R1

- Refer to views::all Behavior of capture vs refer is similar to how views::all works over the expression it is given
- Use wording ‘range adaptor object’ Match current working paper language

§ 1.6 Changes since R0

- Remove customization point objects

Removed views::maybe_has_value and views::maybe_value, instead requiring that the nullable type be dereferenceable and contextually convertible to bool.

- Concept `Nullable`, for exposition

Concept `Nullable`, which is `Readable` and contextually convertible to `bool`

- Capture rvalues by decay copy

Hold a copy when constructing a view over a nullable rvalue.

- Remove `maybe_view` as a specified type

Introduced two exposition types, one safely holding a copy, the other referring to the nullable

§ 2 Before / After Table

Before	After
<pre>{ auto&& opt = possible_value(); if (opt) { // a few dozen lines ... use(*opt); // is *opt OK ? } }</pre>	<pre>for (auto&& opt : views::maybe(possible_value())) { // a few dozen lines ... use(opt); // opt is OK }</pre>
<pre>std::optional o{7}; if (o) { *o = 9; std::cout << "o=" << *o << " prints 9\n"; } std::cout << "o=" << *o << " prints 9\n";</pre>	<pre>std::optional o{7}; for (auto&& i : views::maybe(std::ref(o))) { i = 9; std::cout << "i=" << i << " prints 9\n"; } std::cout << "o=" << *o << " prints 9\n";</pre>

Before	After
<pre> std::vector<int> v{2, 3, 4, 5, 6, 7, 8, 9, 1}; auto test = [](int i) -> std::optional<int> { switch (i) { case 1: case 3: case 7: case 9: return i; default: return {}; } }; auto&& r = v ranges::views::transform(test) ranges::views::filter([](auto x){return bool(x);}) ranges::views::transform([](auto x){return *x;}) ranges::views::transform([](int i) { std::cout << i; return i; }); </pre>	<pre> std::vector<int> v{2, 3, 4, 5, 6, 7, 8, 9, 1}; auto test = [](int i) -> std::optional<int> { switch (i) { case 1: case 3: case 7: case 9: return i; default: return {}; } }; auto&& r = v ranges::views::transform(test) ranges::views::transform(ranges::views::maybe) ranges::views::join ranges::views::transform([](int i) { std::cout << i; return i; }); </pre>

§ 3 Motivation

In writing range transformation it is useful to be able to lift a nullable value into a view that is either empty or contains the value held by the nullable. The adapter `views::single` fills a similar purpose for non-nullable values, lifting a single value into a view, and `views::empty` provides a range of no values of a given type. A `views::maybe` adaptor also allows nullable values to be treated as ranges when it is otherwise undesirable to make them containers, for example `std::optional`.

```

std::vector<std::optional<int>> v{
    std::optional<int>{42},
    std::optional<int>{},
    std::optional<int>{6 * 9}};

auto r = views::join(views::transform(v, views::maybe));

for (auto i : r) {
    std::cout << i; // prints 42 and 54
}

```

In addition to range transformation pipelines, `views::maybe` can be used in range based for loops, allowing the nullable value to not be dereferenced within the body. This is of small value in small examples in contrast to testing the nullable in an if statement, but with longer bodies the dereference is often far away from the test. Often the first line in the body of the if

is naming the dereferenced nullable, and lifting the dereference into the for loop eliminates some boilerplate code, the same way that range based for loops do.

```
{
    auto&& opt = possible_value();
    if (opt) {
        // a few dozen lines ...
        use(*opt); // is *opt OK ?
    }
}

for (auto&& opt : views::maybe(possible_value())) {
    // a few dozen lines ...
    use(opt); // opt is OK
}
```

The view can be on a `std::reference_wrapper`, allowing the underlying nullable to be modified:

```
std::optional o{7};
for (auto&& i : views::maybe(std::ref(o))) {
    i = 9;
    std::cout << "i=" << i << " prints 9\n";
}
std::cout << "o=" << *o << " prints 9\n";
```

Of course, if the nullable is empty, there is nothing in the view to modify.

```
auto oe = std::optional<int>{};
for (int i : views::maybe(std::ref(oe)))
    std::cout << "i=" << i << '\n'; // does not print
```

Converting an optional type into a view can make APIs that return optional types, such a lookup operations, easier to work with in range pipelines.

```
std::unordered_set<int> set{1, 3, 7, 9};

auto flt = [=](int i) -> std::optional<int> {
    if (set.contains(i))
        return i;
    else
        return {};
};

for (auto i : ranges::iota_view{1, 10} | ranges::views::transform(flt)) {
    for (auto j : views::maybe(i)) {
        for (auto k : ranges::iota_view(0, j))
            std::cout << '\a';
        std::cout << '\n';
    }
}

// Produce 1 ring, 3 rings, 7 rings, and 9 rings
```

§ 4 Proposal

Add a range adaptor object `views::maybe`, returning a view over a nullable object, capturing by value temporary nullables. A *nullable* object is one that is both contextually convertible to `bool` and for which the type produced by dereferencing is an equality preserving object. Non void pointers, `std::optional`, and the proposed expected [P0323R9] types all model *nullable*. Function pointers do not, as functions are not objects. Iterators do not generally model *nullable*, as they are not required to be contextually convertible to `bool`.

§ 5 Design

The basis of the design is to hybridize `views::single` and `views::empty`. If the underlying object claims to hold a value, as determined by checking if the object when converted to `bool` is true, begin and end of the view are equivalent to the address of the held value within the underlying object and one past the underlying object. If the underlying object does not have a value, begin and end return `nullptr`.

§ 6 Implementation

A publically available implementation at https://github.com/steve-downey/view_maybe based on the Ranges implementation in `cmcstl2` at <https://github.com/CaseyCarter/cmcstl2>. There are no particular implementation difficulties or tricks. The declarations are essentially what is quoted in the Wording section and the implementations are described as *Effects*.

[Compiler Explorer Link to Before/After Examples](#)

§ 7 LEWG Attention

~~Call LEWG's attention to the use of `ptrdiff_t` as the return type of `size` (which is consistent with `single_view`). The author has a weak preference for a signed type here, but a strong preference for consistency with other Range types: `single_view` now uses `size_t`.~~

Call LEWG's attention to removing the conditional `noexcept` constructors, consistent with `single_view`.

§ 8 Wording

§ 8.1 Synopsis

Modify 24.2 Header synopsis

```
// [range.maybe], maybe view
template<copy_constructible T>
    requires see below
    class maybe_view;

namespace views { inline constexpr unspecified maybe = unspecified; }
```

§ 8.2 Maybe View [\[range.maybe\]](#)

§ 8.2.1 Overview

- 1 `maybe_view` produces a view over a *nullable* that is either empty if the *nullable* is empty, or provides access to the contents of the *nullable* object.
- 2 The name `views::maybe` denotes a customization point object ([customization.point.object]). For some subexpression `E`, the expression `views::maybe(E)` is expression-equivalent to:
 - `maybe_view{E}`, the view specified below, if the expression is well formed, where `decay-copy(E)` is moved into the `maybe_view`
 - otherwise `views::maybe(E)` is ill-formed.

[Note: Whenever `views::maybe(E)` is a valid expression, it is a prvalue whose type models `view`. — end note]

- 3 [*Example*:

```
optional o{4};
maybe_view m{o};
for (int i : m)
    cout << i;           // prints 4
```

— *end example*]

§ 8.2.2 Concept *nullable*

- 1 Types that:
 - are contextually convertible to `bool`
 - are dereferenceable
 - have `const` references which are dereferenceable
 - the `iter_reference_t` of the type and the `iter_reference_t` of the `const` type, will :
 - satisfy `is_lvalue_reference`
 - satisfy `is_object` when the reference is removed
 - for `const` pointers to the referred to types, satisfy `convertible_to`
 - or are a `reference_wrapper` around a type that satisfies *nullable*
- model the exposition only *nullable* concept
- 2 Given a value `i` of type `I`, `I` models *nullable* only if the expression `*i` is equality-preserving. [Note: The expression `*i` is required to be valid via the exposition-only *nullable* concept). — end note]
 - 3 For convenience, the exposition-only *is-reference-wrapper-v* is used below.

```
// For Exposition
template <typename T>
struct is_reference_wrapper : false_type {};
```

```

template <typename T>
struct is_reference_wrapper<reference_wrapper<T>> : true_type {};

template <typename T>
inline constexpr bool is_reference_wrapper_v
    = is_reference_wrapper<T>::value;

// For Exposition
template <class Ref, class ConstRef>
concept readable_references =
    is_lvalue_reference_v<Ref> &&
    is_object_v<remove_reference_t<Ref>> &&
    is_lvalue_reference_v<ConstRef> &&
    is_object_v<remove_reference_t<ConstRef>> &&
    convertible_to<add_pointer_t<ConstRef>,
                  const remove_reference_t<Ref>*>;

template <class T>
concept nullable =
    is_object_v<T> &&
    requires(T& t, const T& ct) {
        bool(ct); // Contextually bool
        *t; // T& is deferenceable
        *ct; // const T& is deferenceable
    }
    && readable_references<iter_reference_t<T>,          // Ref
                          iter_reference_t<const T>>; // ConstRef

template <class T>
concept wrapped_nullable =
    is-reference-wrapper-v<T>
    && nullable<typename T::type>;

```

§ 8.2.3 Class template *maybe_view*

```

namespace std::ranges {
    template <copy_constructible Maybe>
        requires (nullable<Maybe> || wrapped_nullable<Maybe>)
    class maybe_view : public view_interface<maybe_view<Maybe>> {
    private:
        using T = /* see below */
            semiregular_box<Maybe> value_; // exposition only (see 24.7.2
            [range.semi.wrap])

    public:
        constexpr maybe_view() = default;
        constexpr explicit maybe_view(Maybe const& maybe);

        constexpr explicit maybe_view(Maybe&& maybe);

        template<class... Args>
            requires constructible_from<Maybe, Args...>
        constexpr maybe_view(in_place_t, Args&&... args);

        constexpr T*      begin() noexcept;
        constexpr const T* begin() const noexcept;
        constexpr T*      end() noexcept;
        constexpr const T* end() const noexcept;
    };
}

```

```
constexpr size_t size() const noexcept;

constexpr T* data() noexcept;
constexpr const T* data() const noexcept;
};
}
```

```
// For Exposition
using T = remove_reference_t<iter_reference_t<typename
    unwrap_reference_t<Maybe>>>;
```

```
constexpr explicit maybe_view(Maybe const& maybe);
```

- 1 *Effects*: Initializes value_ with maybe.

```
constexpr explicit maybe_view(Maybe&& maybe);
```

- 2 *Effects*: Initializes value_ with move(maybe).

```
template<class... Args>
constexpr maybe_view(in_place_t, Args&&... args);
```

- 3 *Effects*: Initializes value_ as if by value_{in_place, forward<Args>(args)...}.

```
constexpr T* begin() noexcept;
constexpr const T* begin() const noexcept;
```

- 4 *Effects*: Equivalent to: return data();.

```
constexpr T* end() noexcept;
constexpr const T* end() const noexcept;
```

- 5 *Effects*: Equivalent to: return data() + size();.

```
static constexpr ptrdiff_t size() noexcept;
```

- 6 *Effects*: Equivalent to:

```
if constexpr (is-reference-wrapper-v<Maybe>) {
    return bool(value_.get().get());
} else {
    return bool(value_.get());
}
```



```
constexpr T* data() noexcept;
```

- 7 *Effects*: Equivalent to:

```
Maybe& m = value_.get();
if constexpr (is-reference-wrapper-v<Maybe>) {
    return m.get() ? addressof(*(m.get())) : nullptr;
} else {
    return m ? addressof(*m) : nullptr;
}
```



```
constexpr const T* data() const noexcept;
```

8 *Effects*: Equivalent to:

```
const Maybe& m = value_.get();  
if constexpr (is-reference-wrapper-v<Maybe>) {  
    return m.get() ? addressof(*(m.get())) : nullptr;  
} else {  
    return m ? addressof(*m) : nullptr;  
}
```

§ 9 Impact on the standard

A pure library extension, affecting no other parts of the library or language.

§ 10 References

[N4849] Richard Smith. 2020. Working Draft, Standard for Programming Language C++.

<https://wg21.link/n4849>

[P0323R9] JF Bastien, Vicente Botet. 2019. std::expected.

<https://wg21.link/p0323r9>

[P0896R3] Eric Niebler, Casey Carter, Christopher Di Bella. 2018. The One Ranges Proposal.

<https://wg21.link/p0896r3>